



DR. SOUMYA SENGUPTA

PDF | Institut de Ciències de l'Espai

THEORETICAL ASTROPHYSICS | RADIATIVE TRANSFER | SIMULATION | EXOPLANETS

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RESEARCH EXPERIENCE

Position

Research at Open University Israel (2026 – Present)

Hot-Jupiter Chemistry

Post Doctoral Fellow | ARCO, Israel

- **Project Title:** Studying Chemical Analysis of Hot-Jupiter Atmosphere
- **Principle Investigator:** Prof. Allona Vazan
- **Designation:** Post Doctoral Fellow

Ex-Position

Research at Institut de Ciències de l'Espai (2024 – 2025) [⚡]

Studying Hot-Jupiter Radius Inflation Problem

Post Doctoral Fellow | ICE-CSIC, Spain

- **Project Title:** Study of the Hot-Jupiter Radius inflation problem in the presence of Ohmic heat Dissipation using MESA
- **Principle Investigator:** Prof. Daniele Viganò
- **Designation:** Post Doctoral Fellow

Details of Ph.D.:

Research at Indian Institute of Astrophysics (2017 – 2023)

Research on Exoplanets [⚡]

Research Fellow | IIA

- **Thesis Title:** Characterization of the Atmosphere of Extrasolar Planet
- **Supervisor:** Prof. Sujan Sengupta
- **Current Status:** Thesis defended on July 10, 2023, PhD Degree Certificate Obtained.

Publications:

My Publications at Peer-reviewed Journal(s)

Research on Radiative Transfer [⚡]

Research Fellow

- **Soumya Sengupta**, Manika Singla & Fikret Anli, Effect of Thermal Emission in Isotropic Scattering Atmospheres: An Invariant-Embedding Extension of Chandrasekhar's $H()$ -Function, **Accepted for publication in ApJ**; APJ, 1002.2, 2026
- Daniele Viganò, **Sengupta, Soumya**, Clàudia Soriano-Guerrero, Rosalba Perna, Albert Elias-López, Sandeep Kumar, Taner Akgun Inflated hot Jupiters: inferring average atmospheric velocity via Ohmic models coupled with internal dynamo evolution, 2025, A&A, 701, A8
- **Soumya Sengupta** & Sujan Sengupta, Atmospheric heat redistribution effect on Emission spectra of Hot-Jupiters, 2023, New Astronomy, 100, 101987
- **Sengupta, Soumya**, Atmospheric thermal Emission Effect on Chandrasekhar's Finite Atmosphere Problem, 2022, ApJ, 936, 139
- **Sengupta, Soumya**, Effects of Thermal Emission on Chandrasekhars Semi-infinite Diffuse Reflection Problem, 2021, ApJ, 911, 126

Future Research:

Work in Progress

Research Fellow

- Manika Singla, Arvind Rajpurohit, Kinsuk Acharyya, **Soumya Sengupta**, Effect of metallicity, clouds, and orbital separation on the atmosphere of Hot- Jupiters orbiting M-dwarfs I. Emission and Transmission Spectra, under preparation
- ++ **Soumya Sengupta**++ Neutral and Ion Photochemistry in Planetary Atmospheres, Under preparation.

Other Experience:

Teaching Experience

Post Doctoral Fellow | ICE-CSIC

- Conducted Indian Institute of Astrophysics Summer School 2022 and take tutorial course on Exoplanetary Science.
- Serve as a Science Communicator for couple of years of the Indian Institute of Astrophysics Outreach group.

Exoplanetary atmosphere modeling

- PLATO-ESP2025; Planets Throughout the Habitable Zone, Marseille 23-25 June, 2025

Talk Title: **Hot Jupiter Secrets: What Extreme Exoplanets Teach Us About Alien Climates** 

- Colloquim seminar at the Physical Research Laboratory, Ahmedabad, India March 4, 2025

Talk Title: **Hot Jupiter Exoplanets: The Enigmatic Giants of Astrophysics** 

- **Neutral and Ion Photochemistry in Planetary Atmosphere** 29th July - 2nd August 2024,

Poster Title: **Effect of Atmospheric Flow in Hot Jupiters** 

- **7th Institute of Space Sciences Summer School: Multiwavelength Approach to Exoplanets;** 2-11 July 2024,

Participant & LOC 

- **Exoplanets 5** 6 – 21 June 2024 Leiden, the Netherlands

Participant 

- ESP2024: PLATO Planetary systems- formation to observed Architectures; May 14 - 16, 2024

Participant

- *Invited Talk:* Colloquium talk at the Christ University, Bangalore, April 20, 2023

Talk Title: **Exoplanets -The gateway of human search outside of our Solar System** Link 

- *Invited Talk:* Seminar at Center of Excellence in Space Sciences India, Indian Institute of Science Education and Research, Kolkata, 29th March, 2023 

Talk Title: **Modelling of the Exoplanetary Atmosphere -The Theoretical View** Link 

- *Meeting:* Indo-French CEFIPRA Astronomy Meeting (IFCAM)-III January 9 - 13, 2023, IIA, Bengaluru 

Poster Title: **Thermal Emission & Heat-Redistribution effect on Exoplanetary Atmosphere Hot-Jupiter;** Link 

- *Meeting:* 40th Annual Meeting of the Astronomical Society of India (ASI) March 25 - 29, 2022, IIT, Roorkee 

Poster Title: **Thermal emission effect on Chandrasekhar's diffuse reflection problem**

- *Meeting:* 2021 Sagan Exoplanet Summer Virtual Workshop Circumstellar Disks and Young Planets, July 19-23, 2021, NASA Exoplanet Science Institute, California Institute of Technology, PASADENA, CA 

Poster Title: **Thermal emission Effect on Chandrasekhar's Semi-infinite Diffuse Reflection Problem**

- *Meeting:* The 6th UK Exoplanet community Meeting, April 22, 2021, Birmingham (in Virtual mode)

Contributory Talk Title: **Effects of thermal Emission on Chandrasekhara's Semi-infinite Diffuse Reflection Problem**

- *Meeting:* 38th Annual Meeting of the Astronomical Society of India (ASI) February 13 - 17, 2020, IISER, Tirupati 

Poster Title: **Modelling and Characterization of Exoplanet Atmosphere**

- *Meeting:* Young Astronomer's Meet, September 23 - 27, 2019, Kodaikanal Solar Observatory, India 

Talk Title: **Modelling of Exoplanetary Atmosphere**

- *Meeting:* 5th India International Science Festival to be held in Kolkata from 5th to 8th November, 2019

Poster Title: **Modelling of Exoplanetary Atmosphere**

- **Project Title:** Non-Linear analysis of Dark Energy Model
- **Supervisor:** Dr. Anirban Saha, Department of Physics, WBSU

EDUCATION & TRAINING

Qualification

Education

- May, 2022 – 2023: **Post Doctoral Researcher**, Indian Institute of Astrophysics, Bangalore, India
- 2019 – 2022, **Senior Research Fellow**, Indian Institute of Astrophysics, Bangalore, India
- 2017 – 2019, **Junior Research Fellow**, Indian Institute of Astrophysics, Bangalore, India
- 2014 – 2016, **M.Sc. Physics**, West Bengal State University, West Bengal, India, Percentage: 66.27%
- 2014 – 2016, **B.Sc. Physics**, Rastraguru Surendranath College, West Bengal, India, Percentage: 64.25%

RESEARCH INTEREST

- Magnetic Field in exoplanets
- Exoplanetary atmosphere modeling.
- Developing numerical Radiative Transfer code.
- Retrieve chemical abundance and temperature structure of the planetary atmosphere.

COURSES TAKEN

- Radiative Transfer
- Gravitation and Cosmology
- Non-Linear Dynamics
- Astronomy & Astrophysics
- Numerical Methodology
- Fluid Dynamics

OTHER PUBLICATIONS

- The light from heaven - A story of exoplanets (JAI-AWSAR 2020 Award winning story) [🏆]
- "Errors are beautiful and Gaussian" - Really? [🏆] (Published in IIA-DOOT Magazine)

SKILLS, TOOLS & PLATFORMS

✓ Programming & Scripting languages

Python 

C 

FORTRAN 

✓ Markup languages

LaTeX 

MS Word 

Powerpoint 

OTHER SKILLS:

Radius evolution study using MESA code

Analytical & numerical Radiative Transfer modelling

Atmospheric retrieval

Numerical simulation using Discrete Space Theory

Statistical Analysis of atmospheric parameters

PERSONAL INFORMATIONS

Languages: Bengali , English, Hindi

DOB: 11th February, 1994

Father's Name Mr. Sumit Sengupta

Mother's Name Mrs. Sipra Sengupta

Hobbies: Karate, Poetry, Playing Harmonica, Reading, Hiking.